Exercise Set 7.2: Integration by Parts

Step-by-Step Solutions Guide

MAT120: Integral Calculus

October 17, 2026

Overview & Integration by Parts Formula

The Integration by Parts formula is derived from the product rule for differentiation:

$$\int u\,dv = uv - \int v\,du$$

For definite integrals, the formula is:

$$\int_a^b u\,dv = [uv]_a^b - \int_a^b v\,du$$

The rule of thumb **LIATE** (Logarithmic, Inverse Trigonometric, Algebraic, Trigonometric, Exponential) is used to select u in decreasing order of priority.

Complete Solutions (Problems 1–38)

Problem 1: Evaluate $\int xe^{-2x}\,dx$

Solution: Let $u = x \implies du = dx$, and $dv = e^{-2x}\,dx \implies v = -\frac{1}{2}e^{-2x}$. Applying Integration by Parts:

$$\int xe^{-2x}\,dx = -\frac{1}{2}xe^{-2x} - \int \left(-\frac{1}{2}e^{-2x}\right) dx = -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + C = -\frac{1}{4}(2x+1)e^{-2x} + C$$

Problem 2: Evaluate $\int xe^{4x}\,dx$

Solution: Let $u = x \implies du = dx$, and $dv = e^{4x}\,dx \implies v = \frac{1}{4}e^{4x}$.

$$\int xe^{4x}\,dx = \frac{1}{4}xe^{4x} - \int \frac{1}{4}e^{4x}\,dx = \frac{1}{4}xe^{4x} - \frac{1}{16}e^{4x} + C = \frac{1}{16}(4x-1)e^{4x} + C$$

Problem 3: Evaluate $\int x^2e^x\,dx$

Solution: Using tabular integration (or applying integration by parts twice):

1. $u_1 = x^2 \implies du_1 = 2x\,dx, \quad dv_1 = e^x\,dx \implies v_1 = e^x$

$$\int x^2e^x\,dx = x^2e^x - \int 2xe^x\,dx$$

2. For $\int xe^x\,dx$, let $u_2 = x, dv_2 = e^x\,dx \implies \int xe^x\,dx = xe^x - e^x$

Combining both steps:

$$\int x^2 e^x \, dx = x^2 e^x - 2(xe^x - e^x) + C = (x^2 - 2x + 2)e^x + C$$

Problem 4: Evaluate $\int x^2 e^{-2x} \, dx$

Solution: Apply tabular integration with differentiate column $D = \{x^2, 2x, 2, 0\}$ and integrate column $I = \{e^{-2x}, -\frac{1}{2}e^{-2x}, \frac{1}{4}e^{-2x}, -\frac{1}{8}e^{-2x}\}$:

$$\int x^2 e^{-2x} \, dx = x^2 \left(-\frac{1}{2}e^{-2x}\right) - (2x)\left(\frac{1}{4}e^{-2x}\right) + (2)\left(-\frac{1}{8}e^{-2x}\right) + C = -\frac{1}{4}(2x^2 + 2x + 1)e^{-2x} + C$$

Problem 5: Evaluate $\int x^2 \cos x \, dx$

Solution: Tabular integration: $D = \{x^2, 2x, 2, 0\}$ and $I = \{\cos x, \sin x, -\cos x, -\sin x\}$:

$$\int x^2 \cos x \, dx = x^2 \sin x - (2x)(-\cos x) + 2(-\sin x) + C = x^2 \sin x + 2x \cos x - 2 \sin x + C$$

Problem 6: Evaluate $\int x \cos 2x \, dx$

Solution: Let $u = x \implies du = dx$, $dv = \cos 2x \, dx \implies v = \frac{1}{2}\sin 2x$.

$$\int x \cos 2x \, dx = \frac{1}{2}x \sin 2x - \int \frac{1}{2} \sin 2x \, dx = \frac{1}{2}x \sin 2x + \frac{1}{4} \cos 2x + C$$

Problem 7: Evaluate $\int x \sin 3x \, dx$

Solution: Let $u = x \implies du = dx$, $dv = \sin 3x \, dx \implies v = -\frac{1}{3}\cos 3x$.

$$\int x \sin 3x \, dx = -\frac{1}{3}x \cos 3x - \int \left(-\frac{1}{3} \cos 3x\right) dx = -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + C$$

Problem 8: Evaluate $\int x^2 \sin x \, dx$

Solution: Tabular integration: $D = \{x^2, 2x, 2, 0\}$ and $I = \{\sin x, -\cos x, -\sin x, \cos x\}$:

$$\int x^2 \sin x \, dx = x^2(-\cos x) - (2x)(-\sin x) + 2(\cos x) + C = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

Problem 9: Evaluate $\int x \ln x \, dx$

Solution: By LIATE, let $u = \ln x \implies du = \frac{1}{x} \, dx$, and $dv = x \, dx \implies v = \frac{x^2}{2}$.

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C = \frac{x^2}{4}(2 \ln x - 1) + C$$

Problem 10: Evaluate $\int \sqrt{x} \ln x \, dx$

Solution: Let $u = \ln x \implies du = \frac{1}{x} dx$, and $dv = x^{1/2} dx \implies v = \frac{2}{3}x^{3/2}$.

$$\int \sqrt{x} \ln x \, dx = \frac{2}{3}x^{3/2} \ln x - \int \frac{2}{3}x^{3/2} \cdot \frac{1}{x} \, dx = \frac{2}{3}x^{3/2} \ln x - \frac{2}{3} \int x^{1/2} \, dx = \frac{2}{3}x^{3/2} \ln x - \frac{4}{9}x^{3/2} + C$$

Problem 11: Evaluate $\int (\ln x)^2 \, dx$

Solution: Let $u = (\ln x)^2 \implies du = 2 \ln x \cdot \frac{1}{x} dx$, and $dv = dx \implies v = x$.

$$\int (\ln x)^2 \, dx = x(\ln x)^2 - \int x \cdot \frac{2 \ln x}{x} \, dx = x(\ln x)^2 - 2 \int \ln x \, dx$$

For $\int \ln x \, dx$, let $u = \ln x, dv = dx \implies \int \ln x \, dx = x \ln x - x$. Thus:

$$\int (\ln x)^2 \, dx = x(\ln x)^2 - 2(x \ln x - x) + C = x \left((\ln x)^2 - 2 \ln x + 2 \right) + C$$

Problem 12: Evaluate $\int \frac{\ln x}{\sqrt{x}} \, dx$

Solution: Let $u = \ln x \implies du = \frac{1}{x} dx$, and $dv = x^{-1/2} dx \implies v = 2\sqrt{x}$.

$$\int \frac{\ln x}{\sqrt{x}} \, dx = 2\sqrt{x} \ln x - \int 2\sqrt{x} \cdot \frac{1}{x} \, dx = 2\sqrt{x} \ln x - 2 \int x^{-1/2} \, dx = 2\sqrt{x} \ln x - 4\sqrt{x} + C = 2\sqrt{x}(\ln x - 2) + C$$

Problem 13: Evaluate $\int \ln(3x - 2) \, dx$

Solution: Let $u = \ln(3x - 2) \implies du = \frac{3}{3x-2} dx$, and $dv = dx \implies v = x$.

$$\int \ln(3x - 2) \, dx = x \ln(3x - 2) - \int \frac{3x}{3x - 2} \, dx$$

Rewrite $\frac{3x}{3x-2} = 1 + \frac{2}{3x-2}$:

$$\int \frac{3x}{3x - 2} \, dx = \int \left(1 + \frac{2}{3x - 2} \right) dx = x + \frac{2}{3} \ln |3x - 2|$$

Combining terms:

$$\int \ln(3x - 2) \, dx = x \ln(3x - 2) - x - \frac{2}{3} \ln |3x - 2| + C = \left(x - \frac{2}{3} \right) \ln(3x - 2) - x + C$$

Problem 14: Evaluate $\int \ln(x^2 + 16) \, dx$

Solution: Let $u = \ln(x^2 + 16) \implies du = \frac{2x}{x^2+16} dx$, and $dv = dx \implies v = x$.

$$\int \ln(x^2 + 16) \, dx = x \ln(x^2 + 16) - \int \frac{2x^2}{x^2 + 16} \, dx$$

Since $\frac{2x^2}{x^2+16} = 2 - \frac{32}{x^2+16}$:

$$\int \frac{2x^2}{x^2 + 16} \, dx = 2x - 32 \cdot \frac{1}{4} \tan^{-1} \left(\frac{x}{4} \right) = 2x - 8 \tan^{-1} \left(\frac{x}{4} \right)$$

Thus:

$$\int \ln(x^2 + 16) \, dx = x \ln(x^2 + 16) - 2x + 8 \tan^{-1} \left(\frac{x}{4} \right) + C$$

Problem 15: Evaluate $\int \sin^{-1}(2x)\,dx$

Solution: Let $u = \sin^{-1}(2x) \implies du = \frac{2}{\sqrt{1-4x^2}}\,dx$, and $dv = dx \implies v = x$.

$$\int \sin^{-1}(2x)\,dx = x\sin^{-1}(2x) - \int \frac{2x}{\sqrt{1-4x^2}}\,dx$$

For the remaining integral, let $w = 1 - 4x^2 \implies dw = -8x\,dx$:

$$-\int \frac{2x}{\sqrt{1-4x^2}}\,dx = \frac{1}{4}\int w^{-1/2}\,dw = \frac{1}{2}w^{1/2} = \frac{1}{2}\sqrt{1-4x^2}$$

Hence:

$$\int \sin^{-1}(2x)\,dx = x\sin^{-1}(2x) + \frac{1}{2}\sqrt{1-4x^2} + C$$

Problem 16: Evaluate $\int \cos^{-1}(2x)\,dx$

Solution: Let $u = \cos^{-1}(2x) \implies du = -\frac{2}{\sqrt{1-4x^2}}\,dx$, and $dv = dx \implies v = x$.

$$\int \cos^{-1}(2x)\,dx = x\cos^{-1}(2x) - \int \left(-\frac{2x}{\sqrt{1-4x^2}}\right)dx = x\cos^{-1}(2x) - \frac{1}{2}\sqrt{1-4x^2} + C$$

Problem 17: Evaluate $\int \tan^{-1}(3x)\,dx$

Solution: Let $u = \tan^{-1}(3x) \implies du = \frac{3}{1+9x^2}\,dx$, and $dv = dx \implies v = x$.

$$\int \tan^{-1}(3x)\,dx = x\tan^{-1}(3x) - \int \frac{3x}{1+9x^2}\,dx$$

Let $w = 1 + 9x^2 \implies dw = 18x\,dx \implies 3x\,dx = \frac{1}{6}dw$:

$$\int \tan^{-1}(3x)\,dx = x\tan^{-1}(3x) - \frac{1}{6}\ln(1+9x^2) + C$$

Problem 18: Evaluate $\int x\tan^{-1}x\,dx$

Solution: Let $u = \tan^{-1}x \implies du = \frac{1}{1+x^2}\,dx$, and $dv = x\,dx \implies v = \frac{x^2}{2}$.

$$\int x\tan^{-1}x\,dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int \frac{x^2}{1+x^2}\,dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int\left(1 - \frac{1}{1+x^2}\right)dx$$

$$= \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + C = \frac{1}{2}(x^2+1)\tan^{-1}x - \frac{x}{2} + C$$

Problem 19: Evaluate $\int \sin(\ln x)\,dx$

Solution: Let $I = \int \sin(\ln x)\,dx$. Let $u = \sin(\ln x), dv = dx \implies du = \frac{\cos(\ln x)}{x}dx, v = x$.

$$I = x\sin(\ln x) - \int \cos(\ln x)\,dx$$

For $\int \cos(\ln x)\,dx$, let $u = \cos(\ln x), dv = dx \implies du = -\frac{\sin(\ln x)}{x}dx, v = x$.

$$\int \cos(\ln x)\,dx = x\cos(\ln x) + \int \sin(\ln x)\,dx = x\cos(\ln x) + I$$

Substituting back:

$$I = x\sin(\ln x) - x\cos(\ln x) - I \implies 2I = x(\sin(\ln x) - \cos(\ln x))$$

$$I = \frac{x}{2}\left(\sin(\ln x) - \cos(\ln x)\right) + C$$

Problem 20: Evaluate $\int \cos(\ln x)\,dx$

Solution: Similarly to Problem 19, by integrating by parts twice:

$$\int \cos(\ln x)\,dx = \frac{x}{2}\left(\cos(\ln x) + \sin(\ln x)\right) + C$$

Problem 21: Evaluate $\int e^x \sin x\,dx$

Solution: Let $I = \int e^x \sin x\,dx$. 1. Let $u = \sin x, dv = e^x\,dx \implies du = \cos x\,dx, v = e^x$:

$$I = e^x \sin x - \int e^x \cos x\,dx$$

2. For $\int e^x \cos x\,dx$, let $u = \cos x, dv = e^x\,dx \implies du = -\sin x\,dx, v = e^x$:

$$\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I$$

Thus $I = e^x \sin x - e^x \cos x - I \implies 2I = e^x(\sin x - \cos x)$:

$$I = \frac{1}{2}e^x(\sin x - \cos x) + C$$

Problem 22: Evaluate $\int e^{3x} \cos 2x\,dx$

Solution: Let $I = \int e^{3x} \cos 2x\,dx$. 1. Let $u = \cos 2x, dv = e^{3x}\,dx \implies du = -2\sin 2x\,dx, v = \frac{1}{3}e^{3x}$:

$$I = \frac{1}{3}e^{3x}\cos 2x + \frac{2}{3}\int e^{3x}\sin 2x\,dx$$

2. Let $u = \sin 2x, dv = e^{3x}\,dx \implies du = 2\cos 2x\,dx, v = \frac{1}{3}e^{3x}$:

$$\int e^{3x}\sin 2x\,dx = \frac{1}{3}e^{3x}\sin 2x - \frac{2}{3}I$$

Substitute into I:

$$I = \frac{1}{3}e^{3x}\cos 2x + \frac{2}{9}e^{3x}\sin 2x - \frac{4}{9}I \implies \frac{13}{9}I = \frac{e^{3x}}{9}(3\cos 2x + 2\sin 2x)$$

$$I = \frac{e^{3x}}{13}(3\cos 2x + 2\sin 2x) + C$$

Problem 23: Evaluate $\int x \sec^2 x \, dx$

Solution: Let $u = x \implies du = dx$, and $dv = \sec^2 x \, dx \implies v = \tan x$.

$$\int x \sec^2 x \, dx = x \tan x - \int \tan x \, dx = x \tan x - \ln|\sec x| + C = x \tan x + \ln|\cos x| + C$$

Problem 24: Evaluate $\int x \tan^2 x \, dx$

Solution: Use identity $\tan^2 x = \sec^2 x - 1$:

$$\int x \tan^2 x \, dx = \int x(\sec^2 x - 1) \, dx = \int x \sec^2 x \, dx - \int x \, dx$$

Using the result from Problem 23:

$$\int x \tan^2 x \, dx = x \tan x + \ln|\cos x| - \frac{x^2}{2} + C$$

Problem 25: Evaluate $\int x^3 e^{x^2} \, dx$

Solution: Rewrite $x^3 e^{x^2} \, dx = x^2(x e^{x^2} \, dx)$. Let $u = x^2 \implies du = 2x \, dx$, and $dv = x e^{x^2} \, dx \implies v = \frac{1}{2} e^{x^2}$.

$$\int x^3 e^{x^2} \, dx = \frac{1}{2} x^2 e^{x^2} - \int \frac{1}{2} e^{x^2} (2x \, dx) = \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C = \frac{1}{2}(x^2 - 1) e^{x^2} + C$$

Problem 26: Evaluate $\int \frac{xe^x}{(x+1)^2} \, dx$

Solution: Let $u = xe^x \implies du = (xe^x + e^x) \, dx = e^x(x+1) \, dx$. Let $dv = (x+1)^{-2} \, dx \implies v = -\frac{1}{x+1}$.

$$\int \frac{xe^x}{(x+1)^2} \, dx = -\frac{xe^x}{x+1} - \int \left(-\frac{1}{x+1}\right) e^x (x+1) \, dx = -\frac{xe^x}{x+1} + \int e^x \, dx$$

$$= -\frac{xe^x}{x+1} + e^x + C = e^x \left(1 - \frac{x}{x+1}\right) + C = \frac{e^x}{x+1} + C$$

Problem 27: Evaluate $\int_0^2 xe^{3x} \, dx$

Solution: Indefinite integral: $\int xe^{3x} \, dx = \frac{1}{3} xe^{3x} - \frac{1}{9} e^{3x}$. Evaluating from 0 to 2:

$$\left[\frac{1}{3} xe^{3x} - \frac{1}{9} e^{3x}\right]_0^2 = \left(\frac{2}{3} e^6 - \frac{1}{9} e^6\right) - \left(0 - \frac{1}{9}\right) = \frac{5}{9} e^6 + \frac{1}{9}$$

Problem 28: Evaluate $\int_0^1 xe^{-3x} \, dx$

Solution: Indefinite integral: $\int xe^{-3x} \, dx = -\frac{1}{3} xe^{-3x} - \frac{1}{9} e^{-3x}$. Evaluating from 0 to 1:

$$\left[-\frac{1}{3} xe^{-3x} - \frac{1}{9} e^{-3x}\right]_0^1 = \left(-\frac{1}{3} e^{-3} - \frac{1}{9} e^{-3}\right) - \left(0 - \frac{1}{9}\right) = -\frac{4}{9} e^{-3} + \frac{1}{9} = \frac{1 - 4e^{-3}}{9}$$

Problem 29: Evaluate $\int_1^e x^2 \ln x \, dx$

Solution: Let $u = \ln x, dv = x^2\,dx \implies du = \frac{1}{x}dx, v = \frac{x^3}{3}$.

$$\int x^2 \ln x \, dx = \frac{x^3}{3}\ln x - \frac{x^3}{9}$$

Evaluating from 1 to e:

$$\left[\frac{x^3}{3}\ln x - \frac{x^3}{9}\right]_1^e = \left(\frac{e^3}{3} - \frac{e^3}{9}\right) - \left(0 - \frac{1}{9}\right) = \frac{2e^3+1}{9}$$

Problem 30: Evaluate $\int_{\sqrt{e}}^e \frac{\ln x}{x^2}\,dx$

Solution: Let $u = \ln x, dv = x^{-2}\,dx \implies du = \frac{1}{x}dx, v = -x^{-1}$.

$$\int \frac{\ln x}{x^2}\,dx = -\frac{\ln x}{x} - \frac{1}{x}$$

Evaluating from $\sqrt{e}$ to e:

$$\left[-\frac{\ln x + 1}{x}\right]_{\sqrt{e}}^e = \left(-\frac{2}{e}\right) - \left(-\frac{3/2}{\sqrt{e}}\right) = \frac{3}{2\sqrt{e}} - \frac{2}{e}$$

Problem 31: Evaluate $\int_{-1}^1 \ln(x+2)\,dx$

Solution: Indefinite integral: $\int \ln(x+2)\,dx = (x+2)\ln(x+2) - (x+2)$. Evaluating from -1 to 1:

$$[(x+2)\ln(x+2) - x]_{-1}^1 = (3\ln 3 - 1) - (1\ln 1 + 1) = 3\ln 3 - 2 = \ln 27 - 2$$

Problem 32: Evaluate $\int_0^{\sqrt{3}/2} \sin^{-1} x\,dx$

Solution: Indefinite integral: $\int \sin^{-1} x\,dx = x\sin^{-1} x + \sqrt{1-x^2}$. Evaluating from 0 to $\sqrt{3}/2$:

$$\left[x\sin^{-1} x + \sqrt{1-x^2}\right]_0^{\sqrt{3}/2} = \left(\frac{\sqrt{3}}{2}\cdot\frac{\pi}{3} + \sqrt{1-\frac{3}{4}}\right) - (0+1) = \frac{\sqrt{3}\pi}{6} + \frac{1}{2} - 1 = \frac{\sqrt{3}\pi}{6} - \frac{1}{2}$$

Problem 33: Evaluate $\int_2^4 \sec^{-1}\sqrt{\theta}\,d\theta$

Solution: Let $x = \sqrt{\theta} \implies \theta = x^2 \implies d\theta = 2x\,dx$. Limits: $\theta = 2 \implies x = \sqrt{2}$, $\theta = 4 \implies x = 2$.

$$\int 2x\sec^{-1} x\,dx = x^2\sec^{-1} x - \sqrt{x^2-1}$$

Evaluating from $\sqrt{2}$ to 2:

$$\left[x^2\sec^{-1} x - \sqrt{x^2-1}\right]_{\sqrt{2}}^2 = \left(4\cdot\frac{\pi}{3} - \sqrt{3}\right) - \left(2\cdot\frac{\pi}{4} - 1\right) = \frac{4\pi}{3} - \frac{\pi}{2} - \sqrt{3} + 1 = \frac{5\pi}{6} - \sqrt{3} + 1$$

Problem 34: Evaluate $\int_1^2 x \sec^{-1} x \, dx$

Solution: Using the antiderivative $x^2 \frac{\sec^{-1} x}{2} - \frac{1}{2}\sqrt{x^2-1}$: Evaluating from 1 to 2:

$$\left[\frac{x^2 \sec^{-1} x - \sqrt{x^2-1}}{2}\right]_1^2 = \frac{1}{2}\left(\left(4 \cdot \frac{\pi}{3} - \sqrt{3}\right) - (1 \cdot 0 - 0)\right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$$

Problem 35: Evaluate $\int_0^\pi x \sin 2x \, dx$

Solution: Indefinite integral: $-\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x$. Evaluating from 0 to π:

$$\left[-\frac{1}{2}x\cos 2x + \frac{1}{4}\sin 2x\right]_0^\pi = \left(-\frac{\pi}{2}(1) + 0\right) - (0) = -\frac{\pi}{2}$$

Problem 36: Evaluate $\int_0^\pi (x + x\cos x) \, dx$

Solution: Split into $\int_0^\pi x \, dx + \int_0^\pi x\cos x \, dx$. 1. $\int_0^\pi x \, dx = \left[\frac{x^2}{2}\right]_0^\pi = \frac{\pi^2}{2}$. 2. $\int x\cos x \, dx = x\sin x + \cos x \implies [x\sin x + \cos x]_0^\pi = (\pi \cdot 0 - 1) - (0 + 1) = -2$. Thus total integral is:

$$\frac{\pi^2}{2} - 2$$

Problem 37: Evaluate $\int_1^3 \sqrt{x} \tan^{-1}\sqrt{x} \, dx$

Solution: Let $t = \sqrt{x} \implies x = t^2 \implies dx = 2t \, dt$. Limits: $x = 1 \implies t = 1$, $x = 3 \implies t = \sqrt{3}$.

$$\int 2t^2 \tan^{-1} t \, dt = \frac{2}{3}t^3 \tan^{-1} t - \frac{2}{3}\int \frac{t^3}{1+t^2} \, dt = \frac{2}{3}t^3 \tan^{-1} t - \frac{1}{3}t^2 + \frac{1}{3}\ln(1+t^2)$$

Evaluating from 1 to $\sqrt{3}$: Upper limit ($t = \sqrt{3}$):

$$\frac{2}{3}(3\sqrt{3})\left(\frac{\pi}{3}\right) - \frac{3}{3} + \frac{1}{3}\ln 4 = \frac{2\sqrt{3}\pi}{3} - 1 + \frac{2\ln 2}{3}$$

Lower limit ($t = 1$):

$$\frac{2}{3}(1)\left(\frac{\pi}{4}\right) - \frac{1}{3} + \frac{1}{3}\ln 2 = \frac{\pi}{6} - \frac{1}{3} + \frac{\ln 2}{3}$$

Subtracting lower from upper:

$$\left(\frac{2\sqrt{3}\pi}{3} - 1 + \frac{2\ln 2}{3}\right) - \left(\frac{\pi}{6} - \frac{1}{3} + \frac{\ln 2}{3}\right) = \frac{2\sqrt{3}\pi}{3} - \frac{\pi}{6} - \frac{2}{3} + \frac{\ln 2}{3}$$

Problem 38: Evaluate $\int_0^2 \ln(x^2+1) \, dx$

Solution: Antiderivative from Problem 14 with 16 replaced by 1:

$$\int \ln(x^2+1) \, dx = x\ln(x^2+1) - 2x + 2\tan^{-1} x$$

Evaluating from 0 to 2:

$$\left[x\ln(x^2+1) - 2x + 2\tan^{-1} x\right]_0^2 = 2\ln 5 - 4 + 2\tan^{-1}(2)$$